

Programme	B. Sc. Computer Science				
Course Title	Ethical Hacking				
Type of Course	Elective				
Semester	VIII				
Academic Level	400-499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	1. Understanding of the fundamental networking and protocols concepts 2. Familiarity with various operating systems, file systems and basic system administration tasks				
Course Summary	This course provides the skills to identify, analyze, and address security vulnerabilities in systems, networks, and web applications. It aims to learn to perform penetration testing, conduct reconnaissance, exploit vulnerabilities, and maintain access ethically and legally.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO 1	Understand the fundamentals of Ethical Hacking	U	C	Instructor-created exams / Quiz
CO 2	Learn the features of Foot Printing and Reconnaissance	Ap	P	Assignment / Seminar presentations/ Exams
CO 3	Apply the System Hacking methods	Ap	P	Seminar Presentation / Group Tutorial Work/ Viva Voce
CO 4	Understand attacks and type of attacks Apply reasoning with ontologies and rules	U	C	Instructor-created exams / Home Assignments
CO 5	Apply various Penetration Testing methods	Ap	C	Writing assignments/ Exams/ Seminar Presentations
CO 6	Develop theoretical on various types of attack and apply the platforms to explore	Ap	P	Case Study/ Group discussions/ Presentations

* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C)
 # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P)
 Metacognitive Knowledge (M)

Detailed Syllabus:

Module	Unit	Content	Hrs (48+12)	Marks (70)
I	Fundamentals of Ethical Hacking		15	15
	1	Information security overview, Introduction to Hacking, importance of Security – Elements of Security	2	
	2	Hacking Concepts and Hacker Classes - Phases of Hacking Cycle,	3	
	3	Ethical Hacking Tools - Threat and Threat Sources - Malware and Components of Malware -	4	
	4	Types of Malware, Types of Hackers	3	
	5	Common Hacking Methodologies, Benefits and challenges of Ethical Hacking,	3	
II	Foot Printing & Reconnaissance		12	20
	6	Foot Printing & Reconnaissance: Foot printing concepts, Use of foot printing,	2	
	7	information gathering, Types of foot printing, Website Foot printing	2	
	8	Foot printing through Search Engines, Foot Printing through Social Networking sites	2	
	9	Foot Printing tools, Understanding the information gathering process,	2	
	10	Website Foot printing, WHOIS Foot printing,	2	
	11	Network Scanning, Port scanning,	2	
	12	Tools used for the reconnaissance phase		
III	System Hacking		11	20
	13	Password Cracking - Types of Password Attacks	1	
	14	Password Cracking Tools and vulnerabilities	1	
	15	Identity Theft - Social Engineering and tools	2	
	17	Types of attacks and their common prevention mechanisms :	2	
	17	Keystroke Logging, Denial of Service (DoS /DDoS),	2	
	18	Waterhole attack, brute force, phishing and fake WAP, Session Hijacking	3	
IV	Penetration Testing		10	15

	19	Introduction to Penetration Testing, Types of Penetration Testing-	2	
	20	Phases of Penetration Testing,	3	
	21	pen testing, type of pen testing.	3	
	22	Tools of Penetration Testing , Test web applications for vulnerabilities	2	
V	Open Ended Module- Mobile, cloud and IoT Based attacks, Kali Linux		12	
	1	Mobile Platform Attack	3	
	2	Cloud level Attacks and Tools	2	
	3	IoT based attacking Tools	3	
	4	Kali Linux	4	

References

- 1. Michael.T.Simpson, Kent Backman, James.E.Corley, “Hands on Ethical Hacking and Network Defense”, Cengage Learning, 2013
- 2. EC-Council, “Ethical Hacking and Countermeasures Attack Phases”, Cengage Learning
- 3. James S. Tiller, “The Ethical Hack: A Framework for Business Value Penetration Testing”, Auerbach Publications, CRC Press
- Rob Wilson, “Hands-On Ethical Hacking and Network Defense”, Cengage Learning, 2022
- Rafay Boloch, —Ethical Hacking and Penetration Testing Guidel, CRC Press, 2014

Mapping of COs with PSOs and POs :

	PS O1	PSO 2	PSO 3	PSO4	PS O5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	2	1	-	-	1	1						
CO 2	3	-	-	-	1	1						
CO 3	1	3	1	1	2	3						
CO 4	1	-	1	1	2	3						
CO 5	1	-	-	-	2	3						
CO 6	1	2	1	1	3	3						

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium

3	Substantial / High
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Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	End Semester Examinations
CO 1	✓		✓
CO 2	✓		✓
CO 3	✓		✓
CO 4	✓	✓	✓
CO 5	✓	✓	✓
CO 6	✓	✓	